



Linear Programming (LP)

- LP is concerned with the optimization (minimization or maximization) of a **linear function** while satisfying a set of **linear equality and/or inequality constraints** or restrictions.
- Jean Baptiste Joseph Fourier is credited for being the first (1846) to formally state a problem that can be interpreted as a linear programming problem – (finding a solution to a set of linear inequalities).
- LP problem was first conceived by George Dantzig in 1947 while working as Mathematical Advisor to the US Air Force.
- Although a Russian mathematician, LV Kantorovich worked on a similar problem in 1939, his work remained unknown until 1959.
- The term “linear programming” was coined by TC Koopmans (a colleague of Dantzig). Schedules and plans to be implemented by the US Air Force were referred to as programs.



Linear Programming (LP)

- Kantorovich and Koopmans shared the 1975 Nobel Prize in Economic Science
- ...To sum up, Professor Leonid Kantorovich and Tjalling Koopmans, largely independent of one another, have renewed, generalized, and developed methods for the analysis of the classical problem of economics as regards the optimum allocation of scarce resources.



Structure of the LP Problem (Scalar Notation)

$$\text{Maximize} \quad Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n,$$

subject to the restrictions

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m,$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad \dots, \quad x_n \geq 0.$$

How many decision variables are there?

How many constraints are there?



Review of Selected Concepts in Linear Algebra

- Definition of a matrix/vector and its properties
- Identity Matrix
- Elementary row operations
- Solution of a system of linear equations using Gauss-Jordan elimination
- Vector Spaces in R^n
- Addition, Multiplication, scalar multiplication of vectors and matrices
- Linear combinations of vectors

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- Rank of a matrix
- Determinant of a matrix



Structure of the LP Problem (Matrix Notation)

$$\text{Max or Min } Z = cx$$

subject to the restrictions

$$Ax \leq b$$

and

$$x \geq 0$$

What are the dimensions of the matrices?



Components of an LP Problem

- Decision Variables
- Objective Function
- Technological Constraints
- Cost Coefficients
- Technological Coefficients
- Right Hand Side Vector
- Non-negativity Constraints

Maximize $Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n,$

subject to the restrictions

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\leq b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\leq b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\leq b_m,\end{aligned}$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad \dots, \quad x_n \geq 0.$$



Assumptions of LP

- Proportionality - no economies of scale or discounts
- Additivity – total cost is the sum of the individual costs
- Divisibility – decision variables can be divided into fractional levels
- Deterministic

Reference Only



Problem Manipulation

- LP in Standard and Canonical Formats
- Inequalities and Equations
- Non-negativity of decision variables
- Minimization and Maximization problems

Reference Only



Problem Formulation

Reddy Mikks produces both interior and exterior paints from two raw materials, $M1$ and $M2$. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	<i>Exterior paint</i>	<i>Interior paint</i>	
Raw material, $M1$	6	4	24
Raw material, $M2$	1	2	6
Profit per ton (\$1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.



Problem Formulation

x_1 = Tons produced daily of exterior paint

x_2 = Tons produced daily of interior paint

Total profit from exterior paint = $5x_1$ (thousand) dollars

Total profit from interior paint = $4x_2$ (thousand) dollars

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$\left(\begin{array}{c} \text{Usage of a raw material} \\ \text{by both paints} \end{array} \right) \leq \left(\begin{array}{c} \text{Maximum raw material} \\ \text{availability} \end{array} \right)$$

Usage of raw material $M1$ by exterior paint = $6x_1$ tons/day

$$6x_1 + 4x_2 \leq 24 \quad (\text{Raw material } M1)$$

Usage of raw material $M1$ by interior paint = $4x_2$ tons/day

$$x_1 + 2x_2 \leq 6 \quad (\text{Raw material } M2)$$

$$x_2 - x_1 \leq 1 \quad (\text{Market limit})$$

$$x_2 \leq 2 \quad (\text{Demand limit})$$



Problem Formulation

x_1 = Tons produced daily of exterior paint

x_2 = Tons produced daily of interior paint

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



The Stigler Diet

- The **Stigler Diet** (or the Diet Problem) is an optimization problem named for George Stigler, a 1982 Nobel Laureate in economics, who posed the following problem in his paper published in 1945 in the Journal of Farm Economics, "The Cost of Subsistence".
- For a moderately active person weighing 70 kilos, how much of each of 77 foods should be eaten on a daily basis so that the person's intake of nine nutrients will be at least equal to the recommended dietary allowances (RDAs) suggested by the National Research Council, with the cost of the diet being minimal?

Commodity
Wheat Flour (Enriched)
Macaroni
Wheat Cereal (Enriched)
Corn Flakes
Corn Meal
Hominy Grits
Rice
Rolled Oats
White Bread (Enriched)
Whole Wheat Bread
Rye Bread
Pound Cake

Nutrient	Daily Recommended Intake
Calories	3,000 Calories
Protein	70 grams
Calcium	.8 grams
Iron	12 milligrams
Vitamin A	5,000 IU
Thiamine (Vitamin B1)	1.8 milligrams
Riboflavin (Vitamin B2)	2.7 milligrams
Niacin	18 milligrams
Ascorbic Acid (Vitamin C)	75 milligrams



The Diet Problem

- Preliminary weeding reduced the list of eligible foods from 77 to 15, and excludes all meats except liver, all sugars, beverages, and patented cereals.
- Stigler states: "Thereafter the procedure is experimental because there does not appear to be any direct method of finding the minimum of a linear function subject to linear conditions."
- The "optimal solution" (in 1939 dollar values) is shown below.

Food	Annual Quantities	Annual Cost
Wheat Flour	370 lb.	\$13.33
Evaporated Milk	57 cans	\$3.84
Cabbage	111 lb.	\$4.11
Spinach	23 lb.	\$1.85
Dried Navy Beans	285 lb.	\$16.80
Total Annual Cost		\$39.90



The Diet Problem

- In 1947, Jack Laderman used the simplex method (then, a recent invention!) to determine the optimal solution.
- It took 120 man days of nine clerks using desk calculators to derive the optimal solution.
- A typical computer can solve it in less than 300 milliseconds.

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Wheat Flour (Enriched) = 0.029519
Liver (Beef) = 0.001893
Cabbage = 0.011214
Spinach = 0.005008
Navy Beans, Dried = 0.061029

Optimal annual price: \$39.66



The Diet Problem

Wheat Flour (Enriched) = 0.029519

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- Google Chef Anthony Marco took the challenge to create an entrée using these five ingredients and called it *Foie Linéaire à la Stigler* (Linear Optimized Liver à la Stigler).
- Seared calf liver dredged in flour, atop a navy bean purée with marinated cabbage and a spinach pesto.



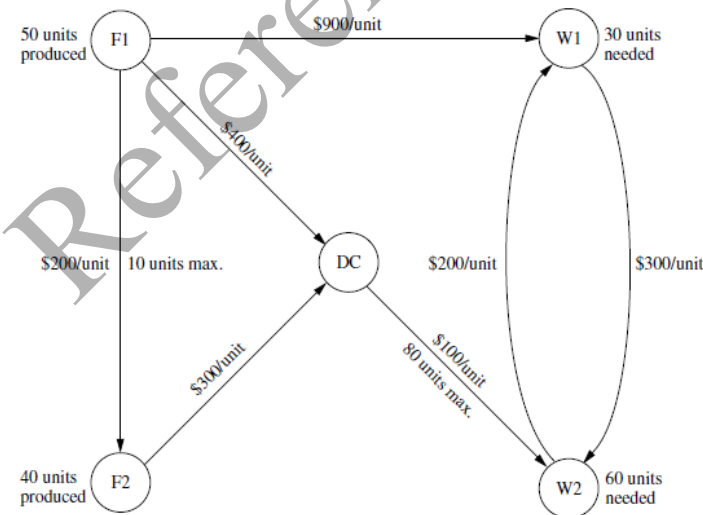
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Problem Formulation

The Problem. The DISTRIBUTION UNLIMITED CO. will be producing the same new product at two different factories, and then the product must be shipped to two warehouses, where either factory can supply either warehouse. The distribution network available for shipping this product is shown in Fig. 3.13, where F1 and F2 are the two factories, W1 and W2 are the two warehouses, and DC is a distribution center. The amounts to be shipped from F1 and F2 are shown to their left, and the amounts to be received at W1 and W2 are shown to their right. Each arrow represents a feasible shipping lane. Thus, F1 can ship directly to W1 and has three possible routes (F1 → DC → W2, F1 → F2 → DC → W2, and F1 → W1 → W2) for shipping to W2. Factory F2 has just one route to W2 (F2 → DC → W2) and one to W1 (F2 → DC → W2 → W1). The cost per unit shipped through each shipping lane is shown next to the arrow. Also shown next to F1 → F2 and DC → W2 are the maximum amounts that can be shipped through these lanes. The other lanes have sufficient shipping capacity to handle everything these factories can send.

The decision to be made concerns how much to ship through each shipping lane. The objective is to minimize the total shipping cost.





Problem Formulation

$x_{F1-F2}, x_{F1-DC}, x_{F1-W1}, x_{F2-DC}, x_{DC-W2}, x_{W1-W2}, x_{W2-W1}$

Number of units to ship from an identified shipping lane

Minimize $Z = 2x_{F1-F2} + 4x_{F1-DC} + 9x_{F1-W1} + 3x_{F2-DC} + x_{DC-W2}$
 $+ 3x_{W1-W2} + 2x_{W2-W1},$

1. Net flow constraints:

$$\begin{aligned} x_{F1-F2} + x_{F1-DC} + x_{F1-W1} &= 50 \text{ (factory 1)} \\ -x_{F1-F2} &+ x_{F2-DC} &= 40 \text{ (factory 2)} \\ -x_{F1-DC} &- x_{F2-DC} + x_{DC-W2} &= 0 \text{ (distribution center)} \\ -x_{F1-W1} &+ x_{W1-W2} - x_{W2-W1} &= -30 \text{ (warehouse 1)} \\ -x_{DC-W2} - x_{W1-W2} + x_{W2-W1} &= -60 \text{ (warehouse 2)} \end{aligned}$$

2. Upper-bound constraints:

$$x_{F1-F2} \leq 10, \quad x_{DC-W2} \leq 80$$

3. Nonnegativity constraints:

$$\begin{aligned} x_{F1-F2} \geq 0, \quad x_{F1-DC} \geq 0, \quad x_{F1-W1} \geq 0, \quad x_{F2-DC} \geq 0, \quad x_{DC-W2} \geq 0, \\ x_{W1-W2} \geq 0, \quad x_{W2-W1} \geq 0. \end{aligned}$$

